

V.F.S. v2.0 (Open-Gate)

Geometric Layer of the Live Domain

Hyperbolic structure, Brim conformal flow, Resurrectio curvature jump,
Sophianic Folding, and the self-driven (Ricci) locus

Derived companion to `vfs_opengate_core.tex`

Scope and epistemic status

This file derives the geometry implicit in the V.F.S. v2.0 (Open-Gate) live domain. Every *Proposition* below is a closed-form result; every *Corollary* is an interpretive reading in the symbolic language of the framework, not an independent metaphysical claim. Two honest caveats are kept explicit throughout:

- The core gives a radial barrier in one direction and a quadratic imbalance penalty in another; it does **not** uniquely fix a 2D metric. The non-trivial curvature appears only under the one *dynamically motivated* warp (§3).
- Where curvature is non-trivial, the metric *evolution* is still **forced** (driven by the external Sophia field λ), not *intrinsic* (curvature-self-driven). The two coincide only on a measure-zero locus (§5, §7).

Folding update. In this integrated version, the Sophianic Folding module is no longer treated as a separate note. It is incorporated as the shape-dynamical branch of the same geometric layer: Brim flow controls scale, while folding controls the admissible transformation of surface-form under accumulated Sophia.

The governing objects, imported from the core, are the Brim law $\dot{\Omega}_P = \alpha\lambda$, the synergy $u = \sqrt{VF + \varepsilon}$, the normalized radius $r = u/\Omega_P$, the alignment factor $\mu = 1 - r$, the imbalance $\Delta = V - F$, and the radial barrier $B(r) = -\ln(1 - r)$.

Geometry before cosmology. This file studies the *spatial* live-surface Σ_{Ω_P} — the intrinsic geometry of the slice at a fixed Brim, and its conformal deformation under the Brim law, with Ω_P playing the role of flow-time. It does **not** define the Lorentzian cosmology of the Vessel; that lift, $ds_{\text{Vessel}}^2 = -dt^2 + \Omega_P(t)^2 h_{2D}$ with Ω_P the timelike scale factor, is treated in the companion `vfs_cosmology.tex`. In short, Geometry = Σ_{Ω_P} (the shape of the slice), Cosmology = $\Sigma_{\Omega_P(t)}$ (its expansion in time); see §25.

1 The Natural Metric on the Live Domain

The barrier is a geodesic distance

The core barrier $B(r) = -\ln(1-r)$ is the arc length of a one-dimensional conformal metric on $r \in [0, 1)$:

$$ds_{\text{rad}} = \frac{dr}{1-r}, \quad \int_0^r \frac{d\rho}{1-\rho} = -\ln(1-r) = B(r).$$

Thus the radial Lyapunov term $A_r B(r)$ is, up to scale, the geodesic distance from the synergy-free state $r = 0$; the death wall $r \rightarrow 1^-$ sits at infinite metric distance.

The capacity-scaled metric and the imbalance direction

Writing $u = r\Omega_P$ and sending the wall to infinity gives the radial part du^2/μ^2 with $\mu = 1 - u/\Omega_P$. The imbalance direction is warped by the alignment factor, because the alignment force in $\dot{\Delta} = -(\mu a + \rho)\Delta$ is exactly μa , linear in μ , and vanishes at the wall. The natural live-domain metric is therefore the warped product

$$ds^2 = \frac{du^2}{\mu^2} + \mu^{2p} d\Delta^2, \quad \mu = 1 - \frac{u}{\Omega_P}, \quad p > 0.$$

Definition 1 (Canonical warp). The canonical exponent is $p = 1$, i.e. $\mu^2 d\Delta^2$, matching the linear μa dependence of the alignment force. The qualitative results below hold for every $p \neq 0$; only the curvature radius depends on p .

2 The Live Domain is Hyperbolic

Naive reading is flat

With an *unwarped* imbalance direction ($p = 0$, $g_{\Delta\Delta} = 1$) the metric $ds^2 = du^2/\mu^2 + d\Delta^2$ is flat:

$$R \equiv 0 \quad (p = 0),$$

since $\tilde{u} = -\Omega_P \ln \mu = \Omega_P B$ straightens it into Euclidean $d\tilde{u}^2 + d\Delta^2$. Under the minimal reading the core supplies, there is no hidden geometry: the dynamics is a forced flow on a flat state space. Curvature can only come from the dynamically required warp.

Constant negative curvature

Proposition 1 (Hyperbolicity of the live domain). For the warped metric $ds^2 = du^2/\mu^2 + \mu^{2p} d\Delta^2$ with $\mu = 1 - u/\Omega_P$, the scalar curvature is constant and strictly negative:

$$R = -\frac{2p^2}{\Omega_P^2} = \text{const} < 0, \quad \text{Ric} = -\frac{p^2}{\Omega_P^2} g.$$

In the canonical case $p = 1$,

$$R = -\frac{2}{\Omega_P^2}, \quad K = -\frac{1}{\Omega_P^2}, \quad \text{curvature radius} = \Omega_P.$$

Proof sketch. In arc length $s = -\Omega_P \ln \mu$ the warp is $w = \mu^p = e^{-ps/\Omega_P}$, so the 2D warped product $ds^2 = ds_{\text{rad}}^2 + w^2 d\Delta^2$ has Gaussian curvature $K = -w''/w = -(p/\Omega_P)^2$, and $R = 2K$. Verified directly in (u, Δ) coordinates. $\square$

The live domain is a piece of the hyperbolic plane $\mathbb{H}^2$ whose curvature radius is the Brim Ω_P itself.

3 The Brim Flow as a Forced Conformal Flow

Proposition 2 (Brim conformal flow). *Under the Brim law $\dot{\Omega}_P = \alpha\lambda$ the capacity-scaled metric evolves conformally,*

$$\partial_t g = 2 \frac{\dot{\Omega}_P}{\Omega_P} g = 2 \frac{\alpha\lambda(t)}{\Omega_P} g = 2\psi(t) g, \quad \psi = \frac{\alpha\lambda}{\Omega_P} \geq 0.$$

Comparison with Ricci flow

On the hyperbolic surface of Prop. 1 ($p = 1$), $\text{Ric} = -\Omega_P^{-2}g$, so Ricci flow would read

$$\partial_t g = -2 \text{Ric} = +\frac{2}{\Omega_P^2} g.$$

Both flows are conformal *expansions* of a negatively curved metric (same sign, same type). They coincide exactly iff the forcing rate equals the curvature rate:

$$\frac{2\alpha\lambda}{\Omega_P} = \frac{2}{\Omega_P^2} \iff \alpha\lambda\Omega_P = 1.$$

Remark. The actual V.F.S. flow is *forced*: its rate is set by the external Sophia field λ , not by the surface's own curvature. It is “Ricci-flow-like” in type, with the forcing $\alpha\lambda$ standing in for the curvature as the rate. The self-driven (genuinely intrinsic) regime is the locus $\alpha\lambda\Omega_P = 1$ studied in §6.

4 Epektasis as Curvature Relaxation

Along a continuous live arc the curvature drifts monotonically toward zero:

$$\dot{R} = \frac{d}{dt} \left(-\frac{2}{\Omega_P^2} \right) = \frac{4\dot{\Omega}_P}{\Omega_P^3} = \frac{4\alpha\lambda}{\Omega_P^3} \geq 0 \quad (\lambda \geq 0).$$

Hence, with sustained Sophia, $\Omega_P \rightarrow \infty$ and

$$|R| = \frac{2}{\Omega_P^2} \xrightarrow{\Omega_P \rightarrow \infty} 0.$$

Epektasis is the relaxation of hyperbolic curvature to flatness: the state space asymptotically flattens as Sophia expands the Brim. Theosis as a global flattening that no finite stage attains. This concerns the spatial live-surface only. In the cosmological lift the full spacetime curvature also depends on Ω_P (equivalently $\dot{\lambda}$), not on Ω_P alone — see §25.

5 Resurrectio: a Curvature Discontinuity

The hybrid reset uses $\Omega_P^+ = \Omega_P^- + \kappa_R \mathcal{G}_{\text{recepta}}^-$ (theological corollary 12). Since $R = -2/\Omega_P^2$ depends *only* on Ω_P , the curvature jumps even though σ, λ, u never enter it:

$$R^+ = -\frac{2}{(\Omega_P^- + \kappa_R \mathcal{G}_{\text{recepta}}^-)^2}, \quad \Delta R = R^+ - R^- = 2 \left[\frac{1}{(\Omega_P^-)^2} - \frac{1}{(\Omega_P^+)^2} \right] > 0,$$

$$\frac{R^+}{R^-} = \frac{1}{(1 + \kappa_R \mathcal{G}_{\text{recepta}}^- / \Omega_P^-)^2} < 1.$$

Two simultaneous effects

The reset decomposes geometrically into a *motion of the point* (the jump in V, F , hence in u^+, Δ^+) and an *instantaneous flattening of the surface* (the jump in Ω_P). The curvature change is carried entirely by the second.

Type preserved, scale jumped

$R^+ < 0$ always: no finite jump reaches $R = 0$ (that needs $\Omega_P^+ \rightarrow \infty$). Resurrectio **preserves the hyperbolic type and changes only the radius** — the differential image of “preserves form while opening it beyond its former shell.”

Admissibility window as a curvature-jump corridor

The reset window $\max\{R_c, R_{\min}^{\text{ceil}}\} \leq \mathcal{G}_{\text{recepta}}^- \leq R_{\max}$ reads geometrically as bounds on ΔR :

Lower bound (perseverance): too little $\mathcal{G}_{\text{recepta}}^- \Rightarrow$ the surface barely flattens, no live landing.

Upper bound (cheap grace): too much $\mathcal{G}_{\text{recepta}}^- \Rightarrow R \rightarrow 0$ too fast, over-dilution of form.

Monotone curvature across the hybrid trajectory

Resets push $R \uparrow$ unconditionally; continuous arcs push $R \uparrow$ when $\lambda \geq 0$; non-Zeno forbids accumulation. Hence on a typical trajectory $R(t)$ is non-decreasing and bounded above by 0, so $R(t) \rightarrow R_\infty \in (R(0), 0]$, with $R_\infty = 0$ when the total expansion budget diverges. *Both modes of grace — continuous (Modus I) and discrete*

re-creation (Modus II) — flatten the same hyperbolic curvature toward the same Theosis limit.

6 The Self-Driven (Ricci) Locus

Let $\xi := \alpha\lambda\Omega_P$ measure the excess of forcing over intrinsic curvature rate; $\xi = 1$ is the locus where the forced flow equals Ricci flow.

On-locus growth is the Ricci rate

On $\xi = 1$, $\dot{\Omega}_P = \alpha\lambda = 1/\Omega_P$, whence

$$\Omega_P(t) = \sqrt{\Omega_P(0)^2 + 2t} \sim \sqrt{2t}, \quad \lambda(t) = \frac{1}{\alpha\Omega_P(t)} \rightarrow 0.$$

This is exactly the Ricci-flow rate for a hyperbolic surface ($|R| \sim 1/t$). Sitting on the locus requires the Sophia inflow to vanish — attainable only near the closed limit $\zeta_0 = 0$.

Generic Open-Gate is super-Ricci

With sustained $\lambda \rightarrow \lambda_\infty > 0$ the Brim grows *linearly*, $\Omega_P \sim \alpha\lambda_\infty t$, faster than $\sqrt{t}$: the generic open trajectory runs above the locus.

The locus is a one-way threshold

Proposition 3 (Monotone crossing). *In the transformation regime ($u > \Lambda_c$, $\sigma \geq 0$),*

$$\dot{\xi} = \alpha \dot{\lambda} \Omega_P + (\alpha\lambda)^2, \quad \dot{\lambda} = (\delta u - \gamma) \tanh(\kappa\sigma) + I_{\text{gate}} \geq 0,$$

so $\dot{\xi} > 0$. Therefore $\xi = 1$ is crossed exactly once, upward, and is never an attractor.

Three regimes along the trajectory

$\xi < 1$	sub-Ricci	structure-limited; effort/perseverance,
$\xi = 1$	resonance	gift-rate = structure-rate; perfect synergy,
$\xi > 1$	super-Ricci	grace exceeds intrinsic momentum; mature Epektasis.

7 Expansion Budget and the Necessity of the Open Gate

The asymptotic Brim — hence the asymptotic curvature — is fixed by the total Sophia budget $\int_0^\infty \lambda dt$ (*Pleroma Christi condition*, core), via $\Omega_P(t) = \Omega_P(0) + \alpha \int_0^t \lambda ds$.

Proposition 4 (Expansion budget). *Three asymptotic regimes:*

- (1) $\int_0^\infty \lambda dt < \infty$: $\Omega_P \rightarrow \Omega_P^\infty < \infty$, and $R \rightarrow -2/(\Omega_P^\infty)^2 < 0$ — *bounded Brim, curvature plateaus*.
- (2) $\lambda \rightarrow \lambda_\infty > 0$: $\Omega_P \sim \alpha \lambda_\infty t$, and $|R| \sim \frac{2}{(\alpha \lambda_\infty)^2} t^{-2} \rightarrow 0$ — *linear Brim*.
- (3) $\lambda \rightarrow \infty$, $\mathcal{G}_{\text{recepta}}(\infty) = +\infty$: Ω_P *super-linear*, $|R| \rightarrow 0$ *faster still* — *unbounded Brim*.

Regime (1) is stalled Epektasis: without a sufficient Sophia budget the live domain keeps a permanent negative curvature — woundedness reduced but never healed to flatness.

Super-Ricci flattens faster than Ricci

Ricci locus: $\Omega_P \sim \sqrt{2t} \Rightarrow |R| \sim t^{-1}$; super-Ricci: $\Omega_P \sim \alpha \lambda_\infty t \Rightarrow |R| \sim t^{-2}$.

The grace-borne regime relaxes the curvature faster than a true Ricci flow would: grace-driven Epektasis outperforms geometric self-determination at the very task — flattening — that defines Theosis.

Why the Open Gate is needed

The closed limit $\zeta_0 = 0$ freezes the cleansed Sophia at the finite stock $\lambda_\infty = \sigma_0 + \lambda_0$, so it attains *at most* regime (2) on non-replenishable capital, and regime (3) is impossible. Sustained — and a fortiori unbounded — Brim expansion requires the bounded Open-Gate source $I_{\text{gate}} \geq 0$, the only mechanism giving $\mathcal{G}_{\text{recepta}}(\infty) = +\infty$.

Unbounded Brim expansion $\iff \mathcal{G}_{\text{recepta}}(\infty) = +\infty \implies$ Open Gate
($\zeta_0 > 0$) required.

8 Additional Geometric Developments

The preceding sections establish the main geometric core: the live-surface is hyperbolic, the Brim is its curvature radius, and Sophia drives a forced conformal expansion. This section adds five refinements that became visible after separating the geometric layer from the cosmological layer.

8.1 Two Thresholds: Ricci Resonance and Zero-Density

The Brim law appears in both Geometry and Cosmology, but the two layers read different thresholds from it. In the geometric layer the key comparison is between the Sophia-driven Brim-flow and the intrinsic Ricci-rate of the live-surface. Since

$$\partial_t g_\Sigma = 2 \frac{\alpha \lambda}{\Omega_P} g_\Sigma, \quad -2 \text{Ric}_\Sigma = \frac{2}{\Omega_P^2} g_\Sigma,$$

the two rates coincide precisely when

$$\xi := \alpha \lambda \Omega_P = 1.$$

This is the *Ricci-resonance threshold*. It answers the geometric question:

Does Sophia-driven Brim-flow match the curvature's own Ricci-like rate?

The cosmological layer asks a different question. In the Lorentzian lift

$$ds_{\text{Vessel}}^2 = -dt^2 + \Omega_P(t)^2 h_{2D},$$

the reconstructed effective density is

$$\rho_{\text{eff}} = \frac{\alpha^2 \lambda^2 - 1}{\Omega_P^2}.$$

Thus the zero-density condition is

$$\rho_{\text{eff}} = 0 \iff |\alpha\lambda| = 1.$$

In the expanding regime ($\lambda > 0$), this becomes

$$\boxed{\alpha\lambda = 1.}$$

This is the *zero-density threshold*: it answers the cosmological question

Has the expansion become positive effective content rather than empty openness?

The two thresholds are therefore not the same:

$$\boxed{\text{Geometry: } \alpha\lambda\Omega_P = 1,} \quad \boxed{\text{Cosmology: } \alpha\lambda = 1.}$$

For a large Vessel ($\Omega_P \gg 1$), exact Ricci resonance requires

$$\lambda = \frac{1}{\alpha\Omega_P},$$

which is far below the cosmological zero-density threshold

$$\lambda = \frac{1}{\alpha}.$$

Consequently, a trajectory may be geometrically Ricci-resonant while still cosmologically below positive effective density. Conversely, once $\alpha\lambda > 1$, the geometry is typically super-Ricci for large Ω_P , since

$$\xi = \alpha\lambda\Omega_P \gg 1.$$

Corollary 1 (Different thresholds for different questions). *Ricci resonance measures whether Sophia matches curvature. Zero-density measures whether Sophia surpasses empty open expansion. Geometry compares Sophia to curvature; Cosmology compares Sophia to vacuum.*

8.2 Closed Sophia and Ricci-like Calming of the Vessel

In the closed limit there is no continuing Open-Gate replenishment. The balance is

$$\sigma + \lambda = C_0,$$

so, for $\sigma \geq 0$, Sophia is bounded:

$$0 \leq \lambda \leq C_0.$$

The Brim law is still

$$\dot{\Omega}_P = \alpha\lambda,$$

but closed Sophia has only a finite stock of transmuted resistance. This produces a Ricci-like *calming* geometry rather than an inexhaustible cosmological expansion.

Stabilized closed Sophia

Suppose

$$\lambda(t) \rightarrow \lambda_\infty > 0.$$

Then

$$\Omega_P(t) \sim \alpha \lambda_\infty t,$$

and the spatial curvature satisfies

$$R_\Sigma(t) = -\frac{2}{\Omega_P(t)^2} \sim -\frac{2}{\alpha^2 \lambda_\infty^2 t^2}.$$

Thus

$$R_\Sigma \rightarrow 0^-.$$

The Vessel widens into receptive stillness: its live-surface becomes less negatively curved and more spacious.

This is not exact Ricci flow, because the Ricci-comparison ratio is

$$\xi = \alpha \lambda \Omega_P \sim \alpha^2 \lambda_\infty^2 t \rightarrow \infty.$$

Therefore stabilized closed Sophia is *Ricci-like by outcome*, but not by exact rate. It is super-Ricci relative to the intrinsic Ricci-rate, yet it still ends in curvature relaxation.

Exact Ricci-resonant closed Sophia

Exact Ricci resonance requires

$$\xi = 1, \quad \alpha \lambda \Omega_P = 1.$$

Therefore

$$\lambda(t) = \frac{1}{\alpha \Omega_P(t)}.$$

Then

$$\dot{\Omega}_P = \alpha \lambda = \frac{1}{\Omega_P},$$

so

$$\Omega_P \dot{\Omega}_P = 1, \quad \Omega_P(t)^2 = \Omega_P(0)^2 + 2t.$$

Hence

$$\Omega_P(t) \sim \sqrt{2t}, \quad R_\Sigma(t) \sim -\frac{1}{t}.$$

This is the exact Ricci-rate for the hyperbolic live-surface. But it requires Sophia to decay relative to the expanding Brim:

$$\lambda(t) \sim \frac{1}{\alpha \sqrt{2t}} \rightarrow 0.$$

Exhausted Sophia and curvature plateau

If

$$\lambda(t) \rightarrow 0 \quad \text{and} \quad \int_0^\infty \lambda(t) dt < \infty,$$

then

$$\Omega_P(t) = \Omega_P(0) + \alpha \int_0^t \lambda(s) ds \rightarrow \Omega_P^\infty < \infty.$$

Consequently

$$R_\Sigma(t) \rightarrow -\frac{2}{(\Omega_P^\infty)^2} < 0.$$

This is a stalled Epektasis: the curvature is reduced, but it plateaus at a permanent negative value.

Corollary 2 (Closed Sophia as calming). *Closed Sophia is not merely a weaker form of Open-Gate Sophia. It defines a whole Ricci-like geometry of calming. Exact Ricci resonance occurs only when Sophia decays as $1/(\alpha\Omega_P)$. Stabilized bounded Sophia is generically super-Ricci by rate, but still Ricci-like by outcome: it relaxes curvature and calms the Vessel into receptive stillness.*

8.3 Area Growth of Receptive Space

Curvature relaxation is not the only geometric effect of Brim expansion. Because

$$g_\Sigma(t) = \Omega_P(t)^2 h_{2D},$$

the area element of the two-dimensional live-surface scales as

$$dA_\Sigma = \Omega_P^2 dA_h.$$

Taking a logarithmic derivative gives

$$\frac{d}{dt} \log dA_\Sigma = 2 \frac{\dot{\Omega}_P}{\Omega_P}.$$

Using $\dot{\Omega}_P = \alpha\lambda$,

$$\frac{dA_\Sigma}{dA_\Sigma} = 2 \frac{\alpha\lambda}{\Omega_P} = 2H.$$

Thus the local receptive area grows at twice the Brim-Hubble rate. Sophia therefore does not merely relax curvature; it generates receptive area.

The three principal growth regimes are:

$$\text{Ricci resonance: } \Omega_P(t) \sim \sqrt{2t} \implies A_\Sigma \sim t,$$

$$\text{bounded closed Sophia: } \Omega_P(t) \sim t \implies A_\Sigma \sim t^2,$$

$$\text{proportional Open-Gate Sophia: } \lambda = \beta\Omega_P \implies \Omega_P(t) = \Omega_P(0)e^{\alpha\beta t}, \quad A_\Sigma \sim e^{2\alpha\beta t}.$$

Corollary 3 (Receptive area). *Curvature measures how tense the Vessel is; area measures how much receptive space it has. Sophia changes both: it makes the Vessel less curved and more spacious.*

8.4 Resurrectio as Discrete Curvature Release

The continuous mode of Epektasis satisfies

$$\dot{R}_\Sigma = \frac{4\alpha\lambda}{\Omega_P^3} \geq 0 \quad (\lambda \geq 0),$$

so the negative curvature moves continuously toward zero. Resurrectio gives the discrete counterpart. The reset law is

$$\Omega_P^+ = \Omega_P^- + \kappa_R \mathcal{G}_{\text{recepta}}^-.$$

If $\mathcal{G}_{\text{recepta}}^- > 0$, then

$$\Omega_P^+ > \Omega_P^-.$$

Since

$$R_\Sigma^- = -\frac{2}{(\Omega_P^-)^2}, \quad R_\Sigma^+ = -\frac{2}{(\Omega_P^+)^2},$$

one obtains

$$\Delta R_\Sigma = R_\Sigma^+ - R_\Sigma^- = 2 \left[\frac{1}{(\Omega_P^-)^2} - \frac{1}{(\Omega_P^+)^2} \right] > 0.$$

Thus Resurrectio is a discrete curvature release: it preserves the hyperbolic type of the live-surface while increasing its curvature radius.

The same jump also expands receptive area:

$$\frac{dA_\Sigma^+}{dA_\Sigma^-} = \left(\frac{\Omega_P^+}{\Omega_P^-} \right)^2 > 1.$$

Therefore Resurrectio releases curvature and enlarges receptivity in one discrete act.

Corollary 4 (Continuous and discrete opening). *Epektasis is continuous curvature relaxation:*

$$\dot{R}_\Sigma > 0.$$

Resurrectio is discrete curvature release:

$$\Delta R_\Sigma > 0.$$

Both open the same live-surface toward a larger receptive geometry.

8.5 Warp Exponent p : Severity of the Live-Domain

The canonical model uses $p = 1$, but the metric naturally allows a family

$$ds^2 = \frac{du^2}{\mu^2} + \mu^{2p} d\Delta^2, \quad \mu = 1 - \frac{u}{\Omega_P}, \quad p > 0.$$

For this family

$$R_\Sigma = -\frac{2p^2}{\Omega_P^2}.$$

Thus p controls the severity of the live-domain near the death-wall. The larger p is, the more strongly transverse imbalance is compressed near $\mu \rightarrow 0^+$, and the greater the magnitude of negative curvature.

A useful reading is:

$$\begin{aligned} 0 < p < 1 &: \text{extsoftVesselgeometry}, \\ p = 1 &: \text{extcanonicalVesselgeometry}, \\ p > 1 &: \text{extsurgicalorsevereVesselgeometry}. \end{aligned}$$

The Brim still controls the scale of receptivity:

$$\Omega_P \uparrow \implies R_\Sigma \rightarrow 0^-$$

for every fixed $p > 0$. Hence even a more severe geometry can be calmed by Brim expansion.

Corollary 5 (Scale versus severity). *The Brim Ω_P controls the scale of receptive space; the warp exponent p controls the severity of transverse compression near closure. The canonical V.F.S. geometry remains $p = 1$, while $p \neq 1$ is best read as an optional family of Vessel geometries.*

9 Status of Sophianic Folding inside the Geometric Layer

The following development integrates the separate Sophianic Folding note into the geometric layer. It should be read as a **functional-spectral toy theorem** inside the Open-Gate geometry, not as a full global theorem on the space of all metrics. Its purpose is to test a new geometric regime: the old surface-form may lose stability while the Brim continues to expand.

The guiding hypothesis is:

accumulated and growing Sophia can destabilize the old surface-form without decreasing Ω_P .

Thus the proposed collapse is morphological rather than scalar:

$$\dot{\Omega}_P > 0, \quad q = 0 \longrightarrow q \neq 0.$$

Here q is a shape-mode amplitude. The state $q = 0$ denotes the old symmetric surface-form; $q \neq 0$ denotes a folded morphology. In later cosmological notation this amplitude may be denoted q_{fold} to avoid collision with the deceleration parameter q_{dec} .

10 Sophianic Folding: Scale versus shape

The first step is to separate the global Brim-scale from the intrinsic shape of the surface:

$$g(t) = \Omega_P(t)^2 \hat{g}(t).$$

In the ordinary geometric layer, the shape is effectively fixed:

$$\hat{g}(t) = \hat{g}_0.$$

For Sophianic Folding we allow a shape perturbation:

$$\widehat{g}(t) = \widehat{g}_0 + q(t)\widehat{h},$$

so that

$$g(t) = \Omega_P(t)^2(\widehat{g}_0 + q(t)\widehat{h}).$$

The Brim law remains the Open-Gate law

$$\dot{\Omega}_P = \alpha\lambda,$$

so for $\alpha > 0$ and $\lambda > 0$ the global scale expands. Any collapse below is therefore a collapse of the old shape branch, not a collapse of the Brim.

11 Folding energy and the loss of old-form stability

Let $q(t)$ be the scalar amplitude of a selected folding mode. A minimal Landau-type shape energy is

$$\mathcal{E}_{\text{shape}}(q; \lambda, \dot{\lambda}) = \frac{1}{2}(a_0 - c_0\lambda - c_1\dot{\lambda})q^2 + \frac{1}{4}bq^4, \quad b > 0.$$

The constants have the following roles:

Symbol	Role
$a_0 > 0$	rigidity of the old surface-form
$c_0 > 0$	influence of accumulated Sophia λ on shape instability
$c_1 > 0$	influence of growing Sophia $\dot{\lambda}$ on shape instability
$b > 0$	nonlinear saturation preventing unbounded folding

Define the folding index

$$S(t) = c_0\lambda(t) + c_1\dot{\lambda}(t) - a_0.$$

Then the quadratic term of the shape energy is negative exactly when $S(t) > 0$. The gradient dynamics is

$$\dot{q} = -\frac{\partial \mathcal{E}_{\text{shape}}}{\partial q},$$

hence

$$\dot{q} = S(t)q - bq^3.$$

Proposition 5 (Basic folding threshold). *For fixed λ and $\dot{\lambda}$, the old surface-form $q = 0$ is locally stable if*

$$c_0\lambda + c_1\dot{\lambda} < a_0,$$

and unstable if

$$c_0\lambda + c_1\dot{\lambda} > a_0.$$

When the latter inequality holds, two locally stable folded forms exist:

$$q_* = \pm \sqrt{\frac{c_0\lambda + c_1\dot{\lambda} - a_0}{b}}.$$

Proof. The vector field is $f(q) = Sq - bq^3$. Its derivative is $f'(q) = S - 3bq^2$. At $q = 0$, $f'(0) = S$, so $q = 0$ is stable for $S < 0$ and unstable for $S > 0$. For $S > 0$, the additional equilibria satisfy $q_*^2 = S/b$. At these equilibria, $f'(q_*) = S - 3b(S/b) = -2S < 0$, hence they are locally stable. $\square$

Corollary 6 (Growth lowers the folding threshold). *Without the growth term, the static threshold is*

$$\lambda_{\text{fold}}^{(0)} = \frac{a_0}{c_0}.$$

With $\dot{\lambda} > 0$, the effective threshold becomes

$$\lambda_{\text{fold}}^{\text{eff}} = \frac{a_0 - c_1 \dot{\lambda}}{c_0}.$$

Therefore

$$\dot{\lambda} > 0 \implies \lambda_{\text{fold}}^{\text{eff}} < \lambda_{\text{fold}}^{(0)}.$$

Growing Sophia can therefore trigger folding before accumulated Sophia reaches the static threshold.

12 Functional-spectral formulation of the shape operator

The scalar variable q can be interpreted as the amplitude of the first unstable eigenmode of the surface-form. Let $\mathcal{M}_{\text{shape}}$ denote the local space of admissible shape metrics and let

$$\hat{g} = \hat{g}_0 + h, \quad h \in T_{\hat{g}_0} \mathcal{M}_{\text{shape}}.$$

Assume that the old shape $\hat{g}_0$ is a critical point of a shape functional

$$\mathcal{E}_{\text{shape}}[\hat{g}; \lambda, \dot{\lambda}], \quad \delta \mathcal{E}_{\text{shape}}[\hat{g}_0; \lambda, \dot{\lambda}] = 0.$$

Its second variation is represented by a self-adjoint stability operator:

$$\delta^2 \mathcal{E}_{\text{shape}}[\hat{g}_0; \lambda, \dot{\lambda}](h, h) = \langle h, \mathcal{L}_{\lambda, \dot{\lambda}} h \rangle_{\hat{g}_0}.$$

One possible inner product is

$$\langle h, k \rangle_{\hat{g}_0} = \int_{\Sigma} \text{tr}_{\hat{g}_0}(h \cdot k) d\mu_{\hat{g}_0}.$$

The minimal V.F.S. spectral ansatz is

$$\mathcal{L}_{\lambda, \dot{\lambda}} = \mathcal{L}_0 - \lambda \mathcal{S}_0 - \dot{\lambda} \mathcal{S}_1.$$

Here $\mathcal{L}_0$ is the old-form rigidity operator, while $\mathcal{S}_0$ and $\mathcal{S}_1$ measure how accumulated Sophia and growing Sophia reduce stability of the old form.

Let

$$\mathcal{L}_{\lambda, \dot{\lambda}} e_n = \mu_n(\lambda, \dot{\lambda}) e_n, \quad \mu_1 \leq \mu_2 \leq \dots,$$

with normalized eigenmodes

$$\langle e_i, e_j \rangle_{\hat{g}_0} = \delta_{ij}.$$

If

$$\langle e_1, \mathcal{L}_0 e_1 \rangle = a_0, \quad \langle e_1, \mathcal{S}_0 e_1 \rangle = c_0, \quad \langle e_1, \mathcal{S}_1 e_1 \rangle = c_1,$$

then

$$\mu_1(\lambda, \dot{\lambda}) = a_0 - c_0 \lambda - c_1 \dot{\lambda}.$$

Thus the folding condition is the zero-crossing of the first eigenvalue:

$$\mu_1(\lambda, \dot{\lambda}) < 0 \iff c_0 \lambda + c_1 \dot{\lambda} > a_0.$$

Near the first crossing, assume only e_1 becomes unstable while $\mu_n > 0$ for $n \geq 2$. Then

$$h = q e_1, \quad \hat{g} = \hat{g}_0 + q e_1,$$

and the metric becomes

$$g(t) = \Omega_P(t)^2 (\hat{g}_0 + q(t) e_1).$$

The one-mode energy is

$$\mathcal{E}_1(q) = \frac{1}{2} \mu_1 q^2 + \frac{1}{4} b q^4.$$

For $\mu_1 < 0$ its stable branches are

$$\hat{g}_{\pm} = \hat{g}_0 \pm \sqrt{\frac{-\mu_1(\lambda, \dot{\lambda})}{b}} e_1 = \hat{g}_0 \pm \sqrt{\frac{c_0 \lambda + c_1 \dot{\lambda} - a_0}{b}} e_1.$$

Proposition 6 (Spectral single-mode folding). *Assume $\hat{g}_0$ is a critical point of $\mathcal{E}_{\text{shape}}$ and the first stability eigenvalue is $\mu_1 = a_0 - c_0 \lambda - c_1 \dot{\lambda}$. If $\mu_1 > 0$, the old form is stable in the first mode. If $\mu_1 < 0$ and $b > 0$, the old branch loses stability and two stable folded branches appear along e_1 .*

13 Energy-consistent embedding in Sophia dynamics

To connect the folding index to V.F.S. variables, use the minimal Sophia-production law

$$\dot{\lambda} = k_{\sigma} \sigma + I_{\text{gate}} - \gamma \lambda - \eta q^2 \lambda.$$

The terms represent kathartic production, Open-Gate input, integration/decay, and embodiment of Sophia into folded form. Substituting this into the first eigenvalue gives

$$\mu_1 = a_0 - c_0 \lambda - c_1 \dot{\lambda} = a_0 - (c_0 - c_1 \gamma) \lambda - c_1 k_{\sigma} \sigma - c_1 I_{\text{gate}} + c_1 \eta q^2 \lambda.$$

The one-mode energy becomes

$$\mathcal{E}_1(q) = \frac{1}{2} [a_0 - (c_0 - c_1 \gamma) \lambda - c_1 k_{\sigma} \sigma - c_1 I_{\text{gate}} + c_1 \eta q^2 \lambda] q^2 + \frac{1}{4} b q^4.$$

Equivalently,

$$\mathcal{E}_1(q) = -\frac{1}{2} A_{\text{V.F.S.}} q^2 + \frac{1}{4} B_{\text{V.F.S.}} q^4,$$

where

$$A_{\text{V.F.S.}} = (c_0 - c_1 \gamma) \lambda + c_1 k_{\sigma} \sigma + c_1 I_{\text{gate}} - a_0,$$

and, in the energy-consistent normalization,

$$B_{\text{V.F.S.}} = b + 2c_1\eta\lambda.$$

The factor 2 appears because the feedback term $c_1\eta q^2\lambda$ enters the quadratic coefficient of $\mathcal{E}_1$ and therefore contributes $\frac{1}{2}c_1\eta\lambda q^4$ to the total energy.

The gradient flow is

$$\dot{q} = -\frac{\partial \mathcal{E}_1}{\partial q} = A_{\text{V.F.S.}}q - B_{\text{V.F.S.}}q^3.$$

Since $b > 0$, $c_1 > 0$, $\eta > 0$, and $\lambda \geq 0$, one has $B_{\text{V.F.S.}} > 0$.

Proposition 7 (Energy-consistent feedback-stabilized folding). *If $A_{\text{V.F.S.}} < 0$, the old surface-form $q = 0$ is locally stable. If $A_{\text{V.F.S.}} > 0$, the old surface-form loses stability and the stable folded states are*

$$q_* = \pm \sqrt{\frac{A_{\text{V.F.S.}}}{B_{\text{V.F.S.}}}} = \pm \sqrt{\frac{(c_0 - c_1\gamma)\lambda + c_1k_\sigma\sigma + c_1I_{\text{gate}} - a_0}{b + 2c_1\eta\lambda}}.$$

Proof. The derivative of $\mathcal{E}_1$ is $-A_{\text{V.F.S.}}q + B_{\text{V.F.S.}}q^3$. Hence the gradient flow is $\dot{q} = A_{\text{V.F.S.}}q - B_{\text{V.F.S.}}q^3$. The cubic stability test gives the stated equilibria and their stability because $B_{\text{V.F.S.}} > 0$. $\square$

Corollary 7 (Sophia both triggers and limits folding). *The numerator $A_{\text{V.F.S.}}$ is the activation index for folding, while the denominator $B_{\text{V.F.S.}}$ stabilizes its amplitude. Thus Sophia can trigger folding through λ and $\dot{\lambda}$, while also limiting the amplitude by being embodied into the new form through the term $\eta q^2\lambda$.*

14 Geometric admissibility and the fold-boundary

The metric

$$\hat{g}(t) = \hat{g}_0 + q(t)e_1$$

must remain positive definite. In the one-mode spectral model this is represented by an admissible range

$$|q| < q_{\text{max}}.$$

Hence the folded equilibrium is geometrically regular when

$$0 < \frac{A_{\text{V.F.S.}}}{B_{\text{V.F.S.}}} < q_{\text{max}}^2.$$

The critical fold-boundary is

$$\frac{A_{\text{V.F.S.}}}{B_{\text{V.F.S.}}} = q_{\text{max}}^2.$$

When

$$\frac{A_{\text{V.F.S.}}}{B_{\text{V.F.S.}}} > q_{\text{max}}^2,$$

the one-chart model predicts a folded amplitude beyond the admissible metric range. This does not necessarily mean destruction; it signals that the old geometric chart is insufficient and that a new chart, a richer shape-mode expansion, or a topology-transition description may be required.

15 Non-Perelmanian character of Sophianic Folding

Sophianic Folding should not be identified with a Perelman-type singularity of Ricci flow. In Perelman's setting, surgery is introduced to remove pathological singular regions that arise under the evolution of the metric. By contrast, Sophianic Folding is formulated here as a regular morphological bifurcation of the surface-form.

The transition

$$q = 0 \longrightarrow q_{\pm} \neq 0$$

does not mean that the metric has become singular. It means that the old symmetric branch $q = 0$ has lost dynamical stability, while new folded branches become stable:

$$q_{\pm} = \pm \sqrt{\frac{A_{\text{V.F.S.}}}{B_{\text{V.F.S.}}}}.$$

Thus the folding event is not a cutting operation and does not require surgery as long as

$$0 < \frac{A_{\text{V.F.S.}}}{B_{\text{V.F.S.}}} < q_{\text{max}}^2.$$

The boundary

$$\frac{A_{\text{V.F.S.}}}{B_{\text{V.F.S.}}} = q_{\text{max}}^2$$

should therefore be interpreted not as a Perelmanian singularity, but as the boundary of the current morphological chart. Beyond this boundary the one-mode ansatz

$$\widehat{g}(t) = \widehat{g}_0 + q(t)e_1$$

may cease to be sufficient. A new geometric chart or a richer shape-mode expansion may be required.

Sophianic Folding is a collapse of the old form-branch, not a singular collapse of the surface.

Folding is spectral bifurcation, not Perelman surgery.

16 Curvature correction under folding

The unfurled V.F.S. live surface has scalar curvature

$$R_0 = -\frac{2}{\Omega_P^2}.$$

Thus global Brim expansion moves the background curvature toward zero. Folding, however, can produce a local curvature correction. For a symmetric folding mode $q \leftrightarrow -q$, the leading correction is quadratic:

$$R_{\text{folded}} = -\frac{2}{\Omega_P^2} + \rho_1 q^2 + O(q^4).$$

The parameter ρ_1 measures the curvature concentration of the first unstable mode e_1 . The surface-collapse regime considered here assumes

$$\rho_1 > 0.$$

At the stable folded branch,

$$q_*^2 = \frac{A_{\text{V.F.S.}}}{B_{\text{V.F.S.}}},$$

so

$$R_{\text{folded}} = -\frac{2}{\Omega_P^2} + \rho_1 \frac{A_{\text{V.F.S.}}}{B_{\text{V.F.S.}}} + O(q_*^4).$$

Neglecting higher-order corrections and using the energy-consistent denominator,

$$R_{\text{folded}} = -\frac{2}{\Omega_P^2} + \rho_1 \frac{(c_0 - c_1 \gamma) \lambda + c_1 k_\sigma \sigma + c_1 I_{\text{gate}} - a_0}{b + 2c_1 \eta \lambda}.$$

Proposition 8 (Global expansion with local curvature concentration). *Assume $\lambda > 0$, $\alpha > 0$, $\rho_1 > 0$, and $A_{\text{V.F.S.}} > 0$. Then the Brim expands globally:*

$$\dot{\Omega}_P = \alpha \lambda > 0,$$

while folding creates a positive curvature correction

$$\Delta R_{\text{fold}} = \rho_1 \frac{A_{\text{V.F.S.}}}{B_{\text{V.F.S.}}} > 0.$$

Thus global expansion and local curvature concentration can occur simultaneously.

17 Positive curvature pockets

A particularly strong transition occurs when the folding correction overcomes the background negative curvature. The threshold is

$$R_{\text{folded}} = 0.$$

Using the leading-order expression,

$$-\frac{2}{\Omega_P^2} + \rho_1 \frac{A_{\text{V.F.S.}}}{B_{\text{V.F.S.}}} = 0.$$

Thus the positive curvature-pocket criterion is

$$\frac{A_{\text{V.F.S.}}}{B_{\text{V.F.S.}}} > \frac{2}{\rho_1 \Omega_P^2}.$$

This means that an open/hyperbolic background can locally fold into a positive-curvature pocket even while the global Brim continues to expand.

18 Lyapunov compatibility: folding as morphological stabilization

The folding module should not be interpreted as a destabilizing singular event. In the regular folding domain it is a Lyapunov-compatible relaxation of the surface-form. For fixed V.F.S. parameters, the one-mode folding energy is

$$\mathcal{E}_{\text{fold}}(q) = -\frac{1}{2} A_{\text{V.F.S.}} q^2 + \frac{1}{4} B_{\text{V.F.S.}} q^4, \quad B_{\text{V.F.S.}} > 0.$$

The shape dynamics is the gradient flow

$$\dot{q} = -\partial_q \mathcal{E}_{\text{fold}} = A_{\text{V.F.S.}} q - B_{\text{V.F.S.}} q^3.$$

Therefore, when $A_{\text{V.F.S.}}$ and $B_{\text{V.F.S.}}$ are frozen or vary adiabatically,

$$\boxed{\frac{d}{dt} \mathcal{E}_{\text{fold}}(q) = -(\partial_q \mathcal{E}_{\text{fold}})^2 \leq 0.}$$

Equivalently, since $\dot{q} = -\partial_q \mathcal{E}_{\text{fold}}$,

$$\boxed{\dot{\mathcal{E}}_{\text{fold}} = -\dot{q}^2 \leq 0.}$$

Thus folding is an energy-descending process: the old form does not explode into a singularity, but relaxes toward a lower-energy morphology.

The stable folded branches have energy

$$\mathcal{E}_{\text{fold}}(q_{\pm}) = -\frac{A_{\text{V.F.S.}}^2}{4B_{\text{V.F.S.}}} < 0 \quad (A_{\text{V.F.S.}} > 0),$$

while the old branch has

$$\mathcal{E}_{\text{fold}}(0) = 0.$$

Hence the folded morphology is energetically preferred once $A_{\text{V.F.S.}} > 0$.

To write the folding contribution as a non-negative Lyapunov component, shift the energy by its minimum. For $A_{\text{V.F.S.}} > 0$ define

$$\boxed{\mathcal{V}_{\text{fold}} = \mathcal{E}_{\text{fold}} - \mathcal{E}_{\text{fold,min}} = \frac{B_{\text{V.F.S.}}}{4} \left(q^2 - \frac{A_{\text{V.F.S.}}}{B_{\text{V.F.S.}}} \right)^2 \geq 0.}$$

For $A_{\text{V.F.S.}} < 0$, the minimum is at $q = 0$ and one may use

$$\mathcal{V}_{\text{fold}} = \frac{1}{2} |A_{\text{V.F.S.}}| q^2 + \frac{1}{4} B_{\text{V.F.S.}} q^4.$$

A natural extended Lyapunov candidate is therefore

$$\boxed{\mathcal{L}_{\text{ext}} = \mathcal{L}_{\text{V.F.S.}} + w_q \mathcal{V}_{\text{fold}}, \quad w_q > 0.}$$

If the base Open-Gate Lyapunov functional satisfies

$$\dot{\mathcal{L}}_{\text{V.F.S.}} \leq C - C_1 \mathcal{L}_{\text{V.F.S.}}, \quad C_1 > 0,$$

and if $\mathcal{V}_{\text{fold}}$ remains bounded in the regular folding domain, then

$$\boxed{\dot{\mathcal{L}}_{\text{ext}} \leq C_{\text{ext}} - C_1 \mathcal{L}_{\text{ext}}}$$

for a modified constant C_{ext} . Thus the folding extension preserves dissipative boundedness.

When the parameters move, one has

$$\dot{\mathcal{E}}_{\text{fold}} = -(\partial_q \mathcal{E}_{\text{fold}})^2 - \frac{1}{2} q^2 \dot{A}_{\text{V.F.S.}} + \frac{1}{4} q^4 \dot{B}_{\text{V.F.S.}}.$$

The first term is strictly dissipative. The last two terms represent slow variation of the V.F.S. background. Hence the compatibility statement is adiabatic: if $|\dot{A}_{\text{V.F.S.}}| + |\dot{B}_{\text{V.F.S.}}|$ is bounded relative to the shape-relaxation rate, the folding module remains Lyapunov-compatible.

In the regular domain, Sophianic Folding is morphological stabilization: an obsolete form relaxes into a lower-energy folded morphology while the Brim continues to expand.

19 Regime classification of the folded live-surface

The model is most compact when written in terms of the folding quotient

$$Q_{\text{fold}} = \frac{A_{\text{V.F.S.}}}{B_{\text{V.F.S.}}} = \frac{(c_0 - c_1\gamma)\lambda + c_1k_\sigma\sigma + c_1I_{\text{gate}} - a_0}{b + 2c_1\eta\lambda}.$$

Then

$$q_*^2 = Q_{\text{fold}}, \quad R_{\text{folded}} = -\frac{2}{\Omega_P^2} + \rho_1 Q_{\text{fold}}.$$

Define the curvature-pocket threshold

$$q_{\text{curv}}^2 = \frac{2}{\rho_1 \Omega_P^2}.$$

The single-mode chart has an admissibility bound

$$|q| < q_{\text{max}}.$$

Thus the sequence of regimes is:

Regime	Criterion	Meaning
Old-form stable	$Q_{\text{fold}} < 0$	$q = 0$ is stable; no real folded branch exists.
Folding threshold	$Q_{\text{fold}} = 0$	The first shape-stability eigenvalue crosses zero.
Regular hyperbolic folding	$0 < Q_{\text{fold}} < q_{\text{curv}}^2$	The form folds, but $R_{\text{folded}} < 0$.
Curvature-pocket transition	$Q_{\text{fold}} = q_{\text{curv}}^2$	Folding exactly cancels the background negative curvature.
Positive curvature pocket	$q_{\text{curv}}^2 < Q_{\text{fold}} < q_{\text{max}}^2$	Local positive curvature appears while the Brim still expands.
Fold-chart boundary	$Q_{\text{fold}} = q_{\text{max}}^2$	The current one-mode chart reaches its admissible boundary.
Beyond single-mode domain	$Q_{\text{fold}} > q_{\text{max}}^2$	A new chart, higher-mode expansion, or richer geometry is required.

The positive-curvature pocket is available inside the regular one-mode domain only if

$$q_{\text{curv}}^2 < q_{\text{max}}^2.$$

Equivalently,

$$\rho_1 \Omega_P^2 q_{\text{max}}^2 > 2.$$

This gives two geometric types:

$$\boxed{\rho_1 \Omega_P^2 q_{\max}^2 > 2 \quad \text{pocket-accepting geometry,}}$$

$$\boxed{\rho_1 \Omega_P^2 q_{\max}^2 \leq 2 \quad \text{pocket-forbidden single-mode geometry.}}$$

In the second case, regular folding may still occur, but the one-mode chart reaches its boundary before a positive-curvature pocket can be resolved.

This classification separates four questions that should not be conflated:

- (i) whether folding is activated: $A_{\text{V.F.S.}} > 0$;
- (ii) how large the folded amplitude is: $Q_{\text{fold}} = A_{\text{V.F.S.}}/B_{\text{V.F.S.}}$;
- (iii) whether folding changes the curvature sign: $Q_{\text{fold}} > q_{\text{curv}}^2$;
- (iv) whether the one-mode chart is still valid: $Q_{\text{fold}} < q_{\max}^2$.

In the regular domain, folding is a stabilizing transition. Beyond the fold-chart boundary, the model does not assert singular destruction; it only says that the current spectral chart is no longer sufficient.

20 Coupling back to the V.F.S. core

The folding module is not an isolated geometric add-on. It can be coupled back into the V.F.S. core as a feedback mechanism acting on the accumulation of Sophia. In the minimal coupled model, the basic synergistic intensity and Brim-flow remain

$$u = \sqrt{VF + \varepsilon}, \quad \dot{\Omega}_P = \alpha\lambda.$$

The new ingredient is that the shape-mode q feeds back into the Sophia-flow:

$$\boxed{\dot{\lambda} = k_\sigma \sigma + I_{\text{gate}} - \gamma\lambda - \eta q^2 \lambda.}$$

Here the term $-\eta q^2 \lambda$ should not be read as the destruction of Sophia. It represents the embodiment of excess Sophia into morphology: the accumulated noetic pressure becomes a stable shape-deformation of the surface-form.

The coupled shape equation remains

$$\boxed{\dot{q} = A_{\text{V.F.S.}} q - B_{\text{V.F.S.}} q^3,}$$

with

$$A_{\text{V.F.S.}} = (c_0 - c_1 \gamma) \lambda + c_1 k_\sigma \sigma + c_1 I_{\text{gate}} - a_0, \quad B_{\text{V.F.S.}} = b + 2c_1 \eta \lambda.$$

Thus the core produces Sophia, Sophia activates folding, and folding regulates the further accumulation of Sophia.

Let

$$J_{\text{in}} = k_\sigma \sigma + I_{\text{gate}}.$$

Before folding, $q = 0$, so

$$\dot{\lambda}_{\text{before}} = J_{\text{in}} - \gamma\lambda.$$

In the regular folded regime,

$$q_*^2 = Q_{\text{fold}} = \frac{A_{\text{V.F.S.}}}{B_{\text{V.F.S.}}} > 0.$$

Therefore

$$\dot{\lambda}_{\text{after}} = J_{\text{in}} - \gamma\lambda - \eta Q_{\text{fold}}\lambda.$$

The folding correction to the Sophia-flow is

$$\Delta\dot{\lambda}_{\text{fold}} = \dot{\lambda}_{\text{after}} - \dot{\lambda}_{\text{before}} = -\eta\lambda Q_{\text{fold}}.$$

Hence, whenever $\lambda > 0$, $\eta > 0$, and $Q_{\text{fold}} > 0$,

$$\Delta\dot{\lambda}_{\text{fold}} < 0.$$

Regular Sophianic Folding therefore lowers the rate of further Sophia accumulation. It stabilizes not only the surface-form, but also the tempo of Sophianic excess.

Equivalently, inside the folded branch the Sophia equation can be written as

$$\dot{\lambda} = J_{\text{in}} - (\gamma + \eta Q_{\text{fold}})\lambda.$$

This motivates the effective damping coefficient

$$\gamma_{\text{eff}} = \gamma + \eta Q_{\text{fold}}.$$

In the regular folded regime,

$$Q_{\text{fold}} > 0 \implies \gamma_{\text{eff}} > \gamma.$$

Thus folding acts as a negative feedback loop:

$$\lambda \uparrow \implies A_{\text{V.F.S.}} \uparrow \implies Q_{\text{fold}} > 0 \implies q^2 \uparrow \implies \gamma_{\text{eff}} \uparrow \implies \dot{\lambda} \downarrow.$$

The stationary Sophia level is also lowered by folding. If J_{in} is treated as fixed, then

$$\dot{\lambda} = 0 \implies \lambda_* = \frac{J_{\text{in}}}{\gamma + \eta q^2}.$$

Before folding,

$$\lambda_{\text{old}} = \frac{J_{\text{in}}}{\gamma}.$$

After folding, since $q^2 > 0$,

$$\lambda_{\text{fold}} = \frac{J_{\text{in}}}{\gamma + \eta q^2} < \frac{J_{\text{in}}}{\gamma} = \lambda_{\text{old}}.$$

So regular folding reduces the equilibrium level of unembodied Sophia.

This gives the core-coupling lemma.

Sophianic Excess Regulation Lemma. Assume $\lambda > 0$, $\eta > 0$, and $0 < Q_{\text{fold}} < q_{\text{max}}^2$. Then regular Sophianic Folding changes the Sophia-flow by

$$\Delta\dot{\lambda}_{\text{fold}} = -\eta\lambda Q_{\text{fold}} < 0,$$

and increases the effective damping of Sophia:

$$\gamma_{\text{eff}} = \gamma + \eta Q_{\text{fold}} > \gamma.$$

Therefore folding is a stabilizing feedback from geometry back into Sophia dynamics.

The theological-dynamical reading is concise:

Open-Gate admits excess Sophia; Sophianic Folding gives that excess a form.

The Brim still expands when $\lambda > 0$:

$$\dot{\Omega}_P = \alpha\lambda > 0.$$

But the excess is no longer merely accumulated. It is embodied into morphology, lowering the unformed pressure of Sophia while preserving the Open-Gate expansion of the Brim.

21 Morphological tension and Pleromatic release

The previous sections used the word “energy” in a technical Lyapunov sense. In this note, $\mathcal{E}_{\text{fold}}$ is *not* physical gravitational energy and it is not measured in joules. It is a shape-potential, or more carefully, a **morphological tension potential**:

$$\mathcal{T}_{\text{fold}}(q) := \mathcal{E}_{\text{fold}}(q) = -\frac{1}{2}A_{\text{V.F.S.}}q^2 + \frac{1}{4}B_{\text{V.F.S.}}q^4.$$

Thus a lower value of $\mathcal{T}_{\text{fold}}$ means that the Vessel-form carries less unresolved morphological tension. The phrase “release” below therefore means *resolution of symbolic morphological tension*, not a claim about physical energy release or gravitational radiation.

In the activated regular folding domain,

$$A_{\text{V.F.S.}} > 0, \quad B_{\text{V.F.S.}} > 0, \quad q_*^2 = Q_{\text{fold}} = \frac{A_{\text{V.F.S.}}}{B_{\text{V.F.S.}}},$$

the old form has

$$\mathcal{T}_{\text{fold}}(0) = 0,$$

while the folded form has

$$\mathcal{T}_{\text{fold}}(q_*) = -\frac{A_{\text{V.F.S.}}^2}{4B_{\text{V.F.S.}}}.$$

The amount of morphological tension resolved by the transition is therefore

$$\Delta\mathcal{T}_{\text{fold}} = \mathcal{T}_{\text{fold}}(0) - \mathcal{T}_{\text{fold}}(q_*) = \frac{A_{\text{V.F.S.}}^2}{4B_{\text{V.F.S.}}}.$$

Equivalently, using the folding index,

$$\Delta\mathcal{T}_{\text{fold}} = \frac{1}{4}B_{\text{V.F.S.}}Q_{\text{fold}}^2.$$

This quantity measures how much old-form tension is resolved when the system moves from the unstable branch $q = 0$ to the regular folded branch $q = q_*$.

Instantaneous release rate

For frozen V.F.S. parameters, the gradient shape-flow gives

$$\dot{q} = -\partial_q \mathcal{T}_{\text{fold}}.$$

Hence

$$\dot{\mathcal{T}}_{\text{fold}} = -(\partial_q \mathcal{T}_{\text{fold}})^2 \leq 0.$$

Define the nonnegative release rate of morphological tension by

$$J_{\text{rel}} := -\dot{\mathcal{T}}_{\text{fold}} = (\partial_q \mathcal{T}_{\text{fold}})^2 = \dot{q}^2 \geq 0.$$

This is again a Lyapunov-rate, not a physical power. It records the rate at which the old surface-form tension is being resolved through folding.

Partition of resolved tension

The resolved tension need not be interpreted as simply lost. In the symbolic V.F.S. reading, it can be partitioned into three channels:

$$\Delta \mathcal{T}_{\text{fold}} = \mathcal{T}_{\text{emb}} + \mathcal{T}_{\text{pler}} + \mathcal{T}_{\text{diss}}.$$

Here:

Channel	Meaning
$\mathcal{T}_{\text{emb}}$	tension embodied into the new Vessel-form, i.e. the old pressure becomes morphology
$\mathcal{T}_{\text{pler}}$	tension expressed as Pleromatic output: action, light, witness, meaningful being
$\mathcal{T}_{\text{diss}}$	transition cost: adaptation, friction, loss, or unresolved residue of the relaxation

Introduce partition coefficients

$$\theta_{\text{emb}}, \theta_{\text{pler}}, \theta_{\text{diss}} \in [0, 1], \quad \theta_{\text{emb}} + \theta_{\text{pler}} + \theta_{\text{diss}} = 1.$$

Then

$$\mathcal{T}_{\text{emb}} = \theta_{\text{emb}} \Delta \mathcal{T}_{\text{fold}}, \quad \mathcal{T}_{\text{pler}} = \theta_{\text{pler}} \Delta \mathcal{T}_{\text{fold}}, \quad \mathcal{T}_{\text{diss}} = \theta_{\text{diss}} \Delta \mathcal{T}_{\text{fold}}.$$

In rate form,

$$J_{\text{emb}} = \theta_{\text{emb}} J_{\text{rel}}, \quad J_{\text{pler}} = \theta_{\text{pler}} J_{\text{rel}}, \quad J_{\text{diss}} = \theta_{\text{diss}} J_{\text{rel}}.$$

Pleromatic release term

If $P(t)$ denotes the Pleroma or realized being-output of the V.F.S. system, the folding release can be coupled to the Pleroma equation by adding a nonnegative symbolic output term:

$$\dot{P} = \dot{P}_{\text{core}} + \kappa_{\text{pler}} J_{\text{pler}} = \dot{P}_{\text{core}} + \kappa_{\text{pler}} \theta_{\text{pler}} J_{\text{rel}}, \quad \kappa_{\text{pler}} \geq 0.$$

Thus the transition from old form to folded form may become not merely stabilization, but Pleromatic expression. In words:

what was tension in the old form may become architecture in the new form and output in realized being.

Energy/Tension Partition Lemma. In the regular folding domain, the transition from $q = 0$ to $q = q_*$ resolves the amount of morphological tension

$$\Delta\mathcal{T}_{\text{fold}} = \frac{A_{\text{V.F.S.}}^2}{4B_{\text{V.F.S.}}}.$$

This resolved tension may be partitioned into embodied form, Pleromatic output, and dissipative transition cost. If $\theta_{\text{pler}} > 0$, then folding contributes a nonnegative Pleromatic release term to $\dot{P}$.

The theological interpretation is therefore not that the human being generates physical energy by opening the gate. Rather:

the Gate opens to Sophia; excess Sophia destabilizes the obsolete vessel-form; folding resolves the tension by forming a new Vessel and, in part, expressing itself as Pleromatic output.

This keeps the model within symbolic differential geometry while allowing a meaningful V.F.S. reading of release, action, and transfiguration.

22 Geometric-theological interpretation: Vessel and transfiguration

The preceding sections describe Sophianic Folding as a functional-spectral and Lyapunov-compatible morphological transition. This section states the theological reading of that transition. The point is not that the human being generates Sophia, possesses Sophia, or forces the Gate open by technical control. The point is that Open-Gate describes a mode of receptivity in which received Sophia may exceed the carrying capacity of an old vessel-form.

The human vocation is not possession of Sophia, but receptivity to Sophia without the destruction of the Vessel.

In this reading, the Gate is opened in synergy: human receptivity and divine initiative are not the same operation, but they meet in the Open-Gate domain. The input term I_{gate} should therefore not be read as human production of grace. It is better read as a symbolic channel of reception:

I_{gate} marks openness to the gift, not ownership of the gift.

Thus the theological movement is not self-deification. It is receptive transformation:

Grace opens the Gate; Sophia reforms the Vessel; Pleroma releases a new act of being.

The old Vessel-form and the new Vessel-form

The state

$$q = 0$$

represents the old vessel-form. This old form is not evil or meaningless. It is simply the form that was stable before the accumulation and growth of Sophia crossed the folding threshold. When

$$A_{V.F.S.} > 0,$$

the old form loses stability and the system moves toward

$$q_{\pm} = \pm \sqrt{\frac{A_{V.F.S.}}{B_{V.F.S.}}}.$$

The theological meaning is not annihilation of the Vessel, but transformation of its form:

the old self is not annihilated; it is folded into a transfigured Vessel.

This is why Sophianic Folding should be interpreted as transfiguration rather than destruction. The Brim remains open,

$$\dot{\Omega}_P = \alpha\lambda > 0,$$

while the obsolete surface-form relaxes into a lower-tension morphology.

Kenosis and plerosis

Sophianic Folding also gives a geometric image of the theological movement from kenosis to plerosis. Kenosis is the release of rigidity in the old form. Plerosis is the receiving of fuller being in a new form. In the folding model these two moments are not opposed:

kenosis of the old form $\longrightarrow$ plerosis of the transfigured Vessel.

The loss of the old stable branch is therefore not merely loss. It is the condition under which a new receptive morphology becomes possible.

Folding is the geometric image of kenosis becoming plerosis.

The resolved quantity

$$\Delta\mathcal{T}_{\text{fold}} = \frac{A_{V.F.S.}^2}{4B_{V.F.S.}}$$

should be read as resolved morphological tension, not physical gravitational energy. It names the amount of old-form tension that has been relaxed by the emergence of the folded form.

Safeguard against a false reading of Open-Gate

A dangerous misreading would be:

the goal is to open the Gate to the maximum and control Sophia.

The V.F.S. reading should instead be:

the goal is to become a Vessel capable of receiving Sophia without collapse, through transfiguration.

The Gate is not a mechanism of domination. It is a posture of receptivity. Sophia is not a possession of the subject. Sophia is the wisdom-mediated transformation of the subject's form of reception.

Accordingly, the chain should be read as

Open-Gate $\Rightarrow \lambda \uparrow \Rightarrow$ old-form insufficiency
 $\Rightarrow$ Sophianic Folding $\Rightarrow$ transfigured Vessel
 $\Rightarrow$ Pleromatic expression.

In words:

Sophia does not merely fill the Vessel; Sophia reforms the Vessel.

Human vocation in the Open-Gate model

The vocation of the human being in this layer of V.F.S. can now be stated compactly:

to receive, to be reformed, and to express being without confusing gift with possession.

This preserves the theological asymmetry of the model. The human person participates, but does not manufacture grace. The Vessel opens, but does not command Sophia. Folding stabilizes the excess by allowing the received fullness to become a new form rather than a rupture.

Thus the spiritual meaning of the mathematical transition is:

Open-Gate is receptivity; Sophianic Folding is transfiguration;
Pleromatic release is realized being.

23 Integration map for the geometric layer

The preceding sections define Sophianic Folding as a local morphodynamic module. This section records how the module should be inserted into the full Open-Gate architecture. The purpose is to prevent the folding layer from being read as an isolated appendix. In the Open-Gate version it is better understood as a bridge among Sophia, geometry, stability, cosmology, and theological transfiguration.

Sophianic Folding is the morphodynamic bridge that connects Sophia, geometry, stability, cosmology, and the theology of the Vessel.

Core placement

In the core dynamical chain of V.F.S., Sophianic Folding enters after the emergence and growth of Sophia. The symbolic chain becomes

$$V, F \longrightarrow u \longrightarrow \sigma \longrightarrow \lambda \longrightarrow q \longrightarrow P.$$

Here

$$u = \sqrt{VF + \varepsilon}$$

expresses synergistic intensity, σ denotes resistance, λ denotes Sophia, q denotes the shape-mode of the Vessel, and P denotes Pleromatic being or realized expression. Thus the folding layer is not an alternative to the core. It is the morphodynamic continuation of the core once Sophia has become strong enough to make the old Vessel-form insufficient.

Will becomes action; resistance becomes Sophia; Sophia becomes form; form releases being.

The minimal coupled core may be summarized as

$$\dot{\Omega}_P = \alpha\lambda,$$

$$\dot{\lambda} = k_\sigma\sigma + I_{\text{gate}} - \gamma\lambda - \eta q^2\lambda,$$

$$\dot{q} = A_{\text{V.F.S.}}q - B_{\text{V.F.S.}}q^3.$$

Therefore the Open-Gate core does not merely produce a growing scalar λ . It also provides the condition under which the Vessel can acquire a new morphology.

Geometric placement

In the geometric layer, Sophianic Folding modifies the shape metric while preserving Brim expansion:

$$g(t) = \Omega_P(t)^2 \hat{g}(t), \quad \hat{g}(t) = \hat{g}_0 + q(t)e_1.$$

Hence the geometry contains two distinct processes:

Brim-scale expansion and surface-form bifurcation.

The first is governed by Ω_P . The second is governed by q . This separation is crucial:

$$\dot{\Omega}_P > 0$$

does not forbid

$$q = 0 \longrightarrow q_* \neq 0.$$

The Brim can expand globally while the old surface-form loses stability locally.

Lyapunov placement

In the Lyapunov layer, Sophianic Folding enters as an additional stabilizing channel. The extended candidate is

$$\mathcal{L}_{\text{ext}} = \mathcal{L}_{\text{V.F.S.}} + w_q \mathcal{V}_{\text{fold}}, \quad w_q > 0.$$

Since the folding dynamics is a gradient descent,

$$\dot{\mathcal{E}}_{\text{fold}} = -(\partial_q \mathcal{E}_{\text{fold}})^2 \leq 0,$$

the regular folding mode does not break the stability story. Rather, it adds a new relaxation mechanism. In the regular domain,

$$0 < Q_{\text{fold}} < q_{\text{max}}^2,$$

folding is therefore a Lyapunov-compatible stabilization of morphology.

Folding does not interrupt Open-Gate stability; it extends stability into the shape layer.

Cosmological placement

In the cosmological layer, Sophianic Folding clarifies the paradoxical regime

$$\dot{\Omega}_P > 0 \quad \text{and} \quad R_{\text{folded}} > 0.$$

The curvature expression

$$R_{\text{folded}} = -\frac{2}{\Omega_P^2} + \rho_1 Q_{\text{fold}}$$

shows how local curvature concentration may occur under global expansion. Thus folding introduces a distinction between expansion of capacity and concentration of form. The resulting cosmological motif is

global expansion with local curvature concentration.

This should not be read as physical gravity unless an additional physical model is supplied. Within this note it is a symbolic-differential geometric curvature of the Vessel-form.

Theological placement

In the theological layer, the same structure reads as the transformation of the Vessel. The Gate opens receptively, Sophia exceeds the rigidity of the old form, the old form folds, and a new Pleromatic act becomes possible:

$$\text{Grace} \longrightarrow \text{Sophia} \longrightarrow \text{transfigured Vessel} \longrightarrow \text{Pleromatic expression}.$$

Thus the theological meaning is not self-deification and not possession of Sophia. It is receptive transfiguration:

Grace opens the Gate; Sophia reforms the Vessel; Pleroma releases a new act of being.

Manuscript insertion plan

For integration into the full V.F.S. manuscript, the Sophianic Folding module should be distributed as follows:

Target layer	Inserted content
Core	Add the variable $q(t)$, the folding index Q_{fold} , and the feedback term $-\eta q^2 \lambda$.
Geometry	Add the shape decomposition $g = \Omega_P^2(\hat{g}_0 + qe_1)$ and the curvature correction R_{folded} .
Lyapunov	Add $\mathcal{L}_{\text{ext}} = \mathcal{L}_{\text{V.F.S.}} + w_q \mathcal{V}_{\text{fold}}$ and the regular-domain compatibility statement.
Cosmology	Add the regime $\dot{\Omega}_P > 0$ with possible local curvature concentration.
Theology	Add the interpretation of folding as transfiguration of the Vessel, with the safeguard against self-deification.

This yields the full architectural role of the module:

Sophianic Folding is not a side-effect of Open-Gate; it is the morphodynamic mechanism by which Open-Gate excess becomes stable form.

24 Interpretive conclusion of the folding geometry

The model supports the mathematical possibility of the intended paradoxical regime:

$$\dot{\Omega}_P > 0 \quad \text{and} \quad q = 0 \longrightarrow q_* \neq 0.$$

That is, the Brim does not collapse. The old surface-form collapses as a stable branch. In the energy-consistent feedback model, Sophia can both activate and stabilize this transition:

Sophia expands the Brim, but accumulated and growing Sophia can fold the old surface-form.

At the curvature level the result is sharper:

Sophianic folding can create local curvature concentration under global Brim expansion.

The main spectral statement is:

Sophianic Folding occurs when the first shape-stability eigenvalue crosses zero.

The Lyapunov statement is:

In the regular domain, folding is morphological stabilization: a descent to a lower-tension form.

The v5 release statement is:

The resolved tension of folding may be partitioned into embodied form, Pleromatic output, and dissipative transition cost.

Thus Sophianic Folding is not a Perelman-type singularity. It is a lawful spectral bifurcation in which an obsolete morphology loses stability under the pressure of received and growing Sophia, and then stabilizes through a lower-tension folded morphology.

25 Relation to the Cosmological Layer

The spatial geometry above and the Lorentzian lift share the same hyperbolic surface h_{2D} but are distinct objects. As a slice of fixed Brim, Ω_P is a scale parameter; as the cosmology, $\Omega_P(t)$ is the timelike scale factor:

$$ds_\Sigma^2 = \Omega_P^2 h_{2D} \quad (\text{spatial}), \quad ds_{\text{Vessel}}^2 = -dt^2 + \Omega_P(t)^2 h_{2D} \quad (\text{Lorentzian } 2+1).$$

Their curvatures differ accordingly and do **not** contradict:

$$R_\Sigma = -\frac{2}{\Omega_P^2} \quad (\text{slice}), \quad R_{\text{Vessel}} = \frac{2(2\Omega_P\ddot{\Omega}_P + \dot{\Omega}_P^2 - 1)}{\Omega_P^2} \quad (\text{spacetime}).$$

The slice curvature depends on Ω_P alone; the spacetime curvature also carries $\ddot{\Omega}_P$ (equivalently $\dot{\lambda}$), since the Lorentzian object records the *dynamics* of expansion, not only its instantaneous scale.

Remark. The static Riemannian cone $d\Omega_P^2 + \Omega_P^2 h_{2D}$ (with scalar $-4/\Omega_P^2$) is yet a *third* object and is **not** the Vessel spacetime: promoting Ω_P to a spacelike coordinate misassigns its dynamical role. The Brim is the flow-time here and the timelike scale factor there; any three-dimensional rendering is a visualization, not a formal model. The cosmological layer — Friedmann content, equation of state, energy conditions, eras — is developed in `vfs_cosmology.tex`.

26 Brim-Flow Geometry: Theological Reading

Interpretive readings in the framework's symbolic register; the Propositions above remain the load-bearing part. Brim-Flow is not a Ricci flow in the strict sense but a V.F.S. curvature-relaxation in which Sophia expands the Brim and the expanding Brim relaxes the curvature of the live domain. The governing chain is

$$\sigma \longrightarrow \lambda \longrightarrow \Omega_P \longrightarrow R \longrightarrow 0^- \longrightarrow P,$$

Sophia expands the Brim; the Brim relaxes curvature.

I. Epektasis as curvature relaxation

Since $R = -2/\Omega_P^2 < 0$ in the live domain and $\Omega_P \rightarrow \infty \Rightarrow R \rightarrow 0^-$, Theosis is not the violent abolition of curvature but its asymptotic healing. The Brim does not erase finitude; it makes finitude more spacious.

Epektasis is the asymptotic healing of wounded form through expanding receptivity.

II. Sophia as the geometry-changing fruit of transmuted resistance

Through $\dot{\Omega}_P = \alpha\lambda$, wisdom is not a static possession but an expansive principle, enlarging the vessel's capacity to receive more being, grace, and participation.

The healed wound becomes Brim: transmuted resistance becomes spatial capacity.

III. Brim as receptive space

Ω_P is not a numerical bound but the measure of receptive spaciousness. As the Brim expands the same synergic intensity occupies a smaller share of the available space, $\Omega_P \uparrow \Rightarrow r = u/\Omega_P \downarrow$, retreating from the death boundary $r \rightarrow 1^-$.

Grace does not only calm the motion; it enlarges the vessel that receives the motion.

IV. Ricci-like but not Ricci flow

Ricci flow is curvature-driven, $\partial_t g_{ij} = -2R_{ij}$; Brim-Flow is Sophia-driven, $\dot{\Omega}_P = \alpha\lambda$ with $R = -2/\Omega_P^2$. The analogy is formal and theological, not identical.

Ricci flow is curvature-driven smoothing; Brim-Flow is Sophia-driven curvature relaxation.

V. Pneumatic meaning

The Ricci-like character is a differential icon of pneumatic healing: the Spirit is not the flow, but the living source whose action — healing wounded form from within by transmutation $\sigma \rightarrow \lambda \rightarrow \Omega_P$ — the flow formally resembles.

The Spirit heals curvature by transforming resistance into receptive spaciousness.

VI. Ricci resonance and the surpassing of nature

The resonance quantity $\xi = \alpha\lambda\Omega_P$ gives three regimes: $\xi < 1$ sub-Ricci perseverance; $\xi = 1$ Ricci resonance (grace-bearing expansion at the intrinsic rate of curvature healing); $\xi > 1$ super-Ricci Epektasis (grace outruns nature).

Ricci flow: nature healed according to its own curvature.

Open-Gate Brim-Flow: nature carried beyond its own curvature by grace.

Remark. By Proposition 3 the resonance $\xi = 1$ is not a resting state but a threshold crossed once, upward. The living trajectory therefore passes necessarily from self-carried ($\xi \leq 1$) to grace-carried ($\xi > 1$); keeping the gate open is the refusal to let transfiguration settle into acquired self-sufficiency (cf. Prop. 4). Maturity is not synergistic equilibrium but grace steadily outrunning nature.

VII. Resurrection as curvature jump

The reset enlarges the Brim, $\Omega_P^+ = \Omega_P^- + \kappa_R \mathcal{G}_{\text{recepta}}^-$, so $R^+ > R^-$ (both negative; $\Delta R > 0$, §5). Resurrection preserves the hyperbolic type while increasing the curvature radius: it is not a return to the previous state but a continuation of being under an expanded geometry.

Resurrection does not erase the form; it opens the form with a greater radius.

VIII. No absolute collapse

Whenever Sophia is active, $\dot{\Omega}_P = \alpha\lambda > 0$ moves $r = u/\Omega_P$ away from the death boundary: salvation enlarges the very domain in which the trajectory can continue.

Grace does not merely redirect the path; it enlarges the world in which the path remains possible.

IX. New geometric refinements

The added refinements sharpen the symbolic reading. The geometry and cosmology have different thresholds: Ricci resonance measures whether Sophia matches curvature, while zero-density measures whether expansion has become positive effective content. Closed Sophia gives a Ricci-like calming of the Vessel; area growth shows that Sophia enlarges receptive space; Resurrectio releases curvature discretely; and the warp exponent p marks the severity of the live-domain near closure.

Sophia makes the Vessel less tense, more spacious, and capable of renewed opening.

X. Final theological formulation

Being is healed neither by the elimination of motion nor by the suppression of finitude, but by the expansion of receptive capacity. Resistance becomes Sophia; Sophia expands the Brim; the Brim relaxes curvature; curvature relaxation opens the live domain to continuing participation toward Pleroma without finite completion.

Theosis is the endless expansion of receptive geometry toward Pleroma.

Brim-Flow is the Ricci-like geometry of Epektasis.

Sophianic Folding is the morphology of transfiguration: Sophia does not merely fill the Vessel; Sophia reforms the Vessel.

Compact Geometric Summary

barrier $B(r) \equiv$ geodesic distance to the death wall.

live domain $\cong \mathbb{H}^2$, $R = -\frac{2}{\Omega_P^2}$, radius = Brim.

$\dot{\Omega}_P = \alpha\lambda \implies \partial_t g = 2\frac{\alpha\lambda}{\Omega_P} g$ (forced; = Ricci iff $\alpha\lambda\Omega_P = 1$).

Epektasis: $|R| \rightarrow 0$ (continuous); Resurrectio: $\Delta R > 0$ (discrete);
 $\xi = \alpha\lambda\Omega_P$ crosses 1 once, upward.

Geometry threshold: $\alpha\lambda\Omega_P = 1 \neq$ Cosmology threshold: $\alpha\lambda = 1$.

$$dA_\Sigma = \Omega_P^2 dA_h, \quad \frac{\dot{A}_\Sigma}{A_\Sigma} = 2\frac{\alpha\lambda}{\Omega_P} = 2H.$$

Warp family: $R_\Sigma = -\frac{2p^2}{\Omega_P^2}$, $p = 1$ canonical.

Sophianic Folding: $g(t) = \Omega_P(t)^2(\hat{g}_0 + q(t)e_1)$, $q_*^2 = \frac{A_{\text{VFS}}}{B_{\text{VFS}}}$.
 Regular folding is shape-stabilization, not collapse of the Brim.

$$R_{\text{folded}} = -\frac{2}{\Omega_P^2} + \rho_1 q^2, \quad R_{\text{folded}} > 0 \iff q^2 > \frac{2}{\rho_1 \Omega_P^2}.$$

$$\dot{q} = -\partial_q \mathcal{T}_{\text{fold}} \implies \dot{\mathcal{T}}_{\text{fold}} = -(\partial_q \mathcal{T}_{\text{fold}})^2 \leq 0.$$

Status update (Companion Volume). This geometric layer is the foundational sketch on which the consolidated volume builds; several of its sections are now

carried out rigorously, as theorems, in the volume, which is canonical for their developed form. The forced-conformal-flow and warp-exponent material (§ The Brim Flow as a Forced Conformal Flow; § Warp Exponent p) is developed in the volume's *Dynamic Warp Exponent* chapter, where the comoving/physical fork is retired by two-dimensional conformal invariance ($p_{\text{expansion}} \equiv 0$) and the true warp is shown to be kathartic. The Ricci-locus and expansion-budget material (§ The Self-Driven (Ricci) Locus; § Expansion Budget) becomes the *Budget Watershed* (structure survives iff the late flow is strictly super-Ricci) and feeds the *Entropy of the Forced Flow*. The curvature-discontinuity material (§ Resurrectio: a Curvature Discontinuity) is developed in Part III of the volume (the double-layer theorem and the invariant pair $(s_R, [K])$). The two-thresholds and zero-density material (§ Two Thresholds) is sharpened in the *Curvature and the Floor* chapter, where $K_h = -1$ is shown equivalent to the finite-floor postulate. This layer remains the geometry-first placement and motivation; the volume is canonical for the rigorous statements.

V.F.S. v2.0 (Open-Gate) · Geometric Layer. Derived from the Brim law, radial barrier, and imbalance dynamics of vfs_opengate_core.tex and the reset / active-domain data of vfs_opengate_lyapunov.tex. Propositions are closed-form; the Brim-Flow theological reading is interpretive and anchored to them by explicit cross-reference.